Luis Victoria

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Work Experience

Software Engineer — Market Data Platform, Capacity & Performance Feb 2023 - Jul 2026
Bloomberg LP *New York, NY*

Engineered data, performance, capacity, scalability, and reliability systems across Bloomberg's Market Data Platform, spanning Feeds and the Ticker Plant. The platform handled >700B ticks/day during extreme volatility while maintaining <6ms median and <250ms p99.9 latency SLAs.

- **Predictive Capacity Forecasting**: Rewrote a fragile legacy log-scraping storage monitor as a two-part Python system: a fleet-wide scanner writing structured per-database storage and machine metadata to a centralized datastore, and a forecasting service extrapolating rolling usage trends 14 days ahead with automated alerting. Delivered in one week during the April 2025 tariff-driven volatility spike, protecting 1,000+ machines from storage-exhaustion/data-loss risk.
- **Performance Regression Detection**: Applied Bayesian Online Changepoint Detection within Argo-orchestrated workflows to statistically flag anomalous times across 30,000+ tick processor instances, cutting remediation time from days to <24 hours.
- **Latency Instrumentation Overhaul**: Authored a C++20 envelope-based timestamping framework tracing tick updates stage-by-stage through the Ticker Plant pipeline with μs resolution, enabling latency regression detection and optimization with $< 10\mu$s overhead.
- **Disaster Recovery Performance Resilience**: Automated load reassignment during data-center failover; benchmarked CPU thread-pinning under combined load, informing Linux cgroups-based CPU prioritization to preserve broadcast continuity while controlling dropped requests.
- **Capacity & Load Management**: Assessed CPU, memory, and storage capacity ahead of onboarding new global exchanges and executed targeted cluster splits to offload overloaded machines, avoiding or eliminating SLO breaches with minimal client disruption.

Quantitative Developer Intern May 2019 - Aug 2019
Farringdon Asset Management *Singapore*

- **Back-testing Enhancements**: Extended the firm's Python backtesting engine to support cryptocurrency analysis, integrating market-data pipelines through CCXT and Binance.
- **Automation with Bloomberg API**: Reduced time to generate client portfolio analysis by 50% for the portfolio management team using information from PORT<GO>.

Education

Business and Computer Science Sep 2018 - Dec 2022
Bachelor of Commerce - Sauder School of Business, University of British Columbia *Vancouver, BC*

- **Relevant Coursework**: Machine Learning & Data Mining, Matrix Algebra, Calculus III, Empirical Economics, Advanced Algorithmic Design, Advanced Relational Databases.
- **Sauder Trading Simulation**: Built a Raspberry Pi arbitrage bot for a primitive non-matching order-book simulation, polling every two minutes and executing crossed bid/ask opportunities; placed 1st of 136 students.